

Math 150A Section 8 and 16
Spring 2025
Practice Exam III Solutions
March 10, 2025
Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write

(a)

$$\int x \cos(x) dx = x \sin(x) + \cos(x) + C$$

TRUE

(b)

$$\sum_{k=1}^4 \frac{1}{k} = \frac{25}{12}$$

TRUE

(c) $1 + 2 + 3 + 4 + \cdots + 100 = 5050$

TRUE

(d) if $\int_0^1 f(x) dx = 1$ then $\int_0^1 3f(x) dx = 3$

TRUE

(e) assuming f and g are continuous

$$\int f(x)g(x)dx = \left(\int f(x)dx \right) \left(\int g(x)dx \right)$$

FALSE

2. (10 points)

(a) Calculate $h'(x)$ for $h(x)$ the function

$$h(x) = \int_{x^3}^x \tan(1/t) dt$$

We use the formula

$$\frac{d}{dx} \int_{u(x)}^{v(x)} f(t) dt = f'(v(x))v'(x) - f'(u(x))u'(x)$$

to get

$$h'(x) = \tan(1/x)(x)' - \tan(1/x^3)(x^3)' = \tan(1/x) - 3x^2 \tan(1/x^3).$$

(b) Calculate $\int_0^3 (\sqrt{9-x^2} + x + 1) dx$ by drawing a picture We break it up into three integrals:

$$\int_0^3 (\sqrt{9-x^2} + x + 1) dx = \int_0^3 \sqrt{9-x^2} dx + \int_0^3 x dx + \int_0^3 1 dx.$$

This consists of a quarter circle of radius 3, a triangle of base 3 and height 3, and a rectangle of base 3 and height 1. Picture not included... The integral is then

$$\int_0^3 (\sqrt{9-x^2} + x + 1) dx = \frac{1}{4}(9\pi) + \frac{1}{2}3^2 + 3.$$

3. (10 points) Use Newton's method to complete the table below and approximate a solution of the equation accurate to four decimal places

$$x + \cos(x) = 0.$$

Newton's method says to use the update equation:

$$x_{n+1} = x_n - \frac{f(x)}{f'(x)},$$

where here $f(x) = x + \cos(x)$. Therefore for our particular situation, the update is

$$x_{n+1} = x_n - \frac{x_n + \cos(x_n)}{1 - \sin(x_n)},$$

Thus

$$x_2 = -1.000000 - \frac{-1.000000 + \cos(-1.000000)}{1 - \sin(-1.000000)} = -0.750364.$$

$$x_3 = -0.750364 - \frac{-0.750364 + \cos(-0.750364)}{1 - \sin(-0.750364)} = -0.739113.$$

$$x_4 = -0.739113 - \frac{-0.739113 + \cos(-0.739113)}{1 - \sin(-0.739113)} = -0.739085.$$

$$x_5 = -0.739085 - \frac{-0.739085 + \cos(-0.739085)}{1 - \sin(-0.739085)} = -0.739085.$$

x_0	0.000000
x_1	-1.000000
x_2	-0.750364
x_3	-0.739113
x_4	-0.739085
x_5	-0.739085

4. (10 points) Compute each of the following integrals. Show all work. Lack of work results in no credit!!

(a) $\int_1^2 u^3 + 3u + 1 du$

$$\begin{aligned}\int_1^2 u^3 + 3u + 1 du &= \left(\frac{1}{4}u^4 + \frac{3}{2}u^2 + u \right) \Big|_1^2 \\ &= \left(\frac{1}{4}2^4 + \frac{3}{2}2^2 + 2 \right) - \left(\frac{1}{4}1^4 + \frac{3}{2}1^2 + 1 \right) = \frac{37}{4}.\end{aligned}$$

(b) $\int (x-1)(x^2+2)dx$

$$\begin{aligned}\int (x-1)(x^2+2)dx &= \int x^3 - x^2 + 2x - 2dx \\ &= \frac{1}{4}x^4 - \frac{1}{3}x^3 + x^2 - 2x + C\end{aligned}$$

(c) $\int_1^8 \frac{3z^3 + \sqrt[3]{z}}{z} dz$

$$\begin{aligned}\int_1^8 \frac{3z^3 + \sqrt[3]{z}}{z} dz &= \int_1^8 3z^2 + z^{-2/3} dz \\ &= (z^3 + 3z^{1/3}) \Big|_1^8 \\ &= (8^3 + 3 \cdot 8^{1/3}) - (1^3 + 3 \cdot 1^{1/3}) = 514\end{aligned}$$

- (d) $\int \csc(t) \cot(t) dt$ Since the derivative of $\csc(t)$ is $-\csc(t) \cot(t)$, we get

$$\int \csc(t) \cot(t) dt = -\csc(t) + C.$$

5. (10 points)

A company wishes to design a fish tank in the shape of a box with an open top, where the length of the base is twice the width. The sides of the tank are glass, which costs 3 dollars per square meter and the base is made out of metal which costs 5 dollars per square meter. If the volume of the tank needs to be 10 cubic meters, what dimensions will minimize the cost of production.

Let the dimensions of the box be x, y, z , where x is the width, y is the length, and z is the height. Then the area of the base is xy , so the cost of the base is

$$C_{\text{base}} = 5xy.$$

The area of the four sides combined is $xz + yz + xz + yz$, so the cost of the sides is

$$C_{\text{sides}} = 3(xz + yz + xz + yz) = 6xz + 6yz.$$

Therefore the total cost is

$$C = 5xy + 6xz + 6yz.$$

Now we know that the length of the base is twice the width, so $y = 2x$. Therefore our cost function is

$$C = 10x^2 + 6xz + 12xz = 10x^2 + 18xz.$$

Finally, the volume needs to be 10 cubic meters, so

$$10 = xyz = 2x^2z.$$

Therefore we see that $z = 5x^{-2}$. Thus our cost is

$$C = 10x^2 + 90x^{-1}.$$

To find the dimensions minimizing this cost, we need to find a critical point. We take the derivative and set it equal to zero, getting

$$20x - 90x^{-2} = 0.$$

Solving this, we get $x = (9/2)^{1/3}$. Plugging this into our equations for y and z , we get that the maximum dimensions are

$$x = (9/2)^{1/3}, \quad y = 2(9/2)^{1/3}, \quad z = 5(2/9)^{2/3}.$$

6. (10 points)

(a) Determine an integral whose value is equal to $\lim_{n \rightarrow \infty} R_n$ where here

$$R_n = \sum_{k=1}^n \sin\left(1 + \frac{3k}{n}\right) \frac{3}{n}.$$

The Riemann sum for the integral $\int_a^b f(x)dx$ is

$$R_n = \sum_{k=1}^n f(x_k) \Delta x, \quad \Delta x = \frac{b-a}{n}, \quad x_k = a + k\Delta x.$$

From this, we can identify $a = 1$, $f(x) = \sin(x)$, and $\Delta x = \frac{3}{n}$. Therefore $\frac{b-a}{n} = \frac{3}{n}$, so that $b - a = 3$ and since $a = 1$ this says $b = 4$. This says the integral we want is $\int_1^4 \sin(x)dx$.

(b) Compute the value of the left Riemann sum L_3 for $f(x) = \cos(x)$ on the interval $[-\pi/3, \pi/3]$. The width of our rectangles is $\Delta x = 2\pi/9$. Therefore using the left endpoints of the rectangles

$$L_3 = \cos(-\pi/3)(2\pi/9) + \cos(-\pi/9)(2\pi/9) + \cos(\pi/9)(2\pi/9).$$

7. (10 points)

Consider the Riemann integral

$$\int_0^2 x^2 + x dx$$

- (a) Determine an explicit expression for the right Riemann sum R_n . The widths of the rectangles are $\Delta x = \frac{2-0}{n} = 2/n$. Therefore $x_k = 0 + k\Delta x = 2k/n$. The right Riemann sum is

$$R_n = \sum_{k=1}^n f(x_k) \Delta x = \sum_{k=1}^n \left(\left(\frac{2k}{n} \right)^2 + \left(\frac{2k}{n} \right) \right) \frac{2}{n}.$$

- (b) Compute Riemann integral by calculating the limit $\lim_{n \rightarrow \infty} R_n$. We can rewrite the above sum as

$$\begin{aligned} R_n &= \sum_{k=1}^n \left(\left(\frac{2k}{n} \right)^2 + \left(\frac{2k}{n} \right) \right) \frac{2}{n} \\ &= \sum_{k=1}^n \frac{8k^2}{n^3} + \frac{4k}{n^2} \\ &= \sum_{k=1}^n \frac{8k^2}{n^3} + \sum_{k=1}^n \frac{4k}{n^2} \\ &= \frac{8}{n^3} \sum_{k=1}^n k^2 + \frac{4}{n^2} \sum_{k=1}^n k = \frac{8}{n^3} \frac{n(n+1)(2n+1)}{6} + \frac{4}{n^2} \frac{n(n+1)}{2}. \end{aligned}$$

Taking the limit, we get

$$\begin{aligned} \lim_{n \rightarrow \infty} R_n &= \lim_{n \rightarrow \infty} \left(\frac{8(2n^3 + 3n^2 + n)}{6n^3} + \frac{4n^2 + 4n}{2n^2} \right) \\ &= \lim_{n \rightarrow \infty} \frac{8(2n^3 + 3n^2 + n)}{6n^3} + \lim_{n \rightarrow \infty} \frac{4n^2 + 4n}{2n^2} \\ &= \frac{16}{6} + \frac{4}{2} = \frac{28}{6}. \end{aligned}$$